

Neural Field Perceiver

Cross-Patient Speech Decoding from Intra-Operative μ ECoG
via Spatially-Grounded Electrode Encoding

Ben Tang | Greg Cogan Lab, Duke University

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Technical design document. **One idea:** each patient’s electrode array is a partial, noisy observation of a shared neural manifold for speech production. We recover this manifold via atlas-grounded VE cross-attention at Brainnetome positions, mapping variable electrode counts into a fixed-size common space. Per-patient diagonal normalization and rigid coordinate correction calibrate each “view.” An autoregressive decoder generates 3-phoneme sequences via beam search over 52 valid CVC/VCV tokens. Training: external chronic ECoG SSL (JEPA with temporal masking, 50–100 h) $\rightarrow$ per-patient supervised fine-tuning. Electrode self-attention before VE cross-attention is a first-class ablation (A_elec_attn).

Claims labeled **supported**, **observed**, or **hypothesis**. Structural assumptions that could be wrong: **red**.

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1 Problem

Each patient’s μ ECoG array is a sparse spatial sample of cortical high-gamma activity (HGA, 70–150 Hz, 200 Hz) during speech. The array sits on left ventral sensorimotor cortex (vSMC), and electrode positions in MNI-152 space are determined post-hoc from surgical reconstruction.

Hardware. LCP thin-film arrays (Dyconex [1]), physically flexible — they conform to cortical curvature. Two configurations:

	128-ch (DBS)	256-ch (craniotomy)
Grid	8×16	12×22
Pitch	1.33 mm	1.72 mm
Footprint	$\sim 11 \times 21$ mm	$\sim 21 \times 38$ mm
Contact diameter	200 μ m	200 μ m

Task. Non-word repetition (intra-operative, immediate — no imposed delay): 52 CVC/VCV tokens, 9 phonemes, 3 positions per token. Output: three 9-way classifications per trial. Full-trial input (default $t_{\min}=-0.2$, $t_{\max}=1.0$ s relative to response onset, pending Phase 0 sweep). Stimulus-to-response delay is 1.1 ± 0.3 s [1], so the auditory stimulus ends ~ 400 ms before $t_{\min}$. Autoregressive decoder generates [phoneme₁, phoneme₂, phoneme₃] sequentially; eliminates MFA dependency.

Cross-patient challenge. Arrays are offset 15–25 mm in MNI space with variable rotation. Channel indices are meaningless across patients. Current LOPO achieves PER=0.762 (S14, grouped-by-token CV) using a per-patient Conv2d read-in that ignores electrode coordinates entirely. 55 architecture/training experiments converge to PER 0.750–0.780, suggesting a systematic barrier.

Constraints.

Constraint	Value	Impact
Sig. electrodes/patient	63–201	Sparse cortical sampling
Cross-patient offset	15–25 mm	Different cortex sampled
Trials/patient	46–178	Small-data regime
Utterance/patient	~ 1 min	SSL has failed
Patients	~ 10	Too few for large models

Why coordinates might help. Per-patient Conv2d learns spatial filters independently — nothing is spatially shared. If two patients sample the same cortical region, the model cannot exploit this. MNI coordinates make spatial correspondence explicit.

2 Idea: Spatially-Grounded Electrode Encoding

2.1 What we assume

Speech motor cortex has **conserved somatotopic organization**: gross ordering (medial→lateral: leg→hand→face) is invariant [3]; functional connectivity is the most conserved feature [5]. Fine structure varies by ~ 15 –25 mm [4]. The spatial map exists; its absolute position varies.

Untested: That this conservation is strong enough, at μ ECoG resolution, to transfer across $N=10$ patients with centimeter-scale positional uncertainty.

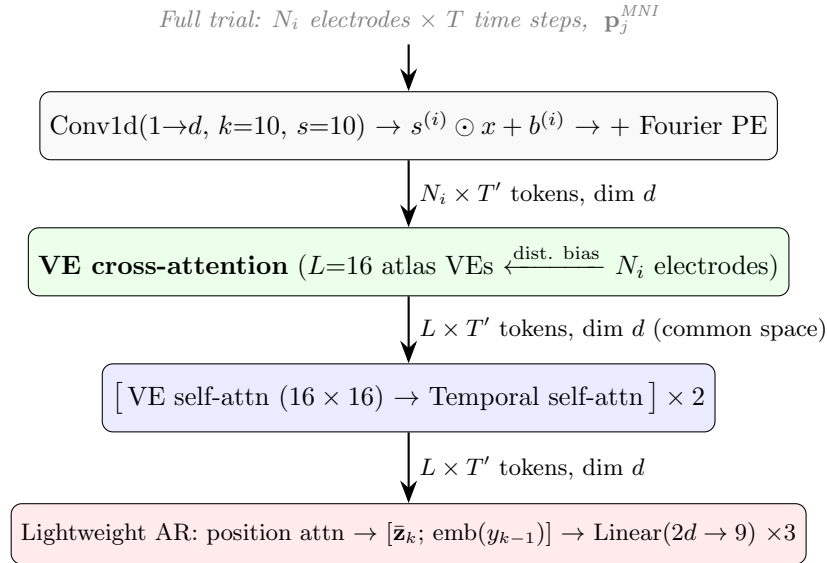
2.2 The core components

The idea has three components, motivated by multi-view reconstruction:

1. **Per-patient calibration.** Diagonal normalization ($s^{(i)} \odot x + b^{(i)}$) absorbs impedance/gain. Learned rigid correction ($\Delta^{(i)}, \omega^{(i)}$) absorbs registration error. Per-patient VE position offsets ($\delta_l^{(i)}$, leashed ≤ 15 mm) adapt atlas positions to individual somatotopy. These are the “camera intrinsics and extrinsics.”
2. **Atlas-grounded virtual electrodes.** $L=16$ query vectors at per-patient-adapted Brainnetome positions (atlas centroid + learned offset) cross-attend to real electrodes with a distance bias, mapping variable-count electrodes into a **common functional space** — the “template mesh.” The atlas ($N=40$ HCP subjects [6]) provides the spatial prior; per-patient offsets (48 params) adapt to individual variation.
3. **Temporal self-attention.** Self-attention among VE tokens across time captures articulatory dynamics. Temporal span masking during SSL (Section 5) teaches the model to predict missing temporal content from context.

Total per-patient params: ~ 182 (128 diagonal + 6 geometric + 48 VE offsets). Far below BIT (6M/pt), Willett (65.8K/day), seegnicant (2K/subject). Details in Sections 3.1–3.3.

3 Architecture



$$\mathcal{L} = \sum_i w^{(i)} \mathcal{L}_{class}^{(i)} + \lambda_{\Delta\omega} E_{\Delta\omega} + \lambda_{sm} E_{smooth}$$

$$\mathcal{L}_{supervised} = \sum_i w^{(i)} \frac{1}{3} \sum_{k=1}^3 \text{FocalCE}(\hat{y}_k, y_k) + \lambda_{\Delta\omega} E_{\Delta\omega} + \lambda_{sm} E_{smooth} \quad (1)$$

$$\mathcal{L}_{\text{SSL}} = \text{JEPA}_{L_1} \text{ with temporal masking, content-aware weighting, } r \sim \text{U}(0, 1) \quad (\text{Section 5.1})$$

(2)

3.1 Electrode tokens

Each electrode is a point in MNI space with a scalar HGA time series. Tokenization:

$$\text{token}_j(t') = \underbrace{s^{(i)} \odot \text{Conv1d}(x_j; k=10, s=10)_{t'} + b^{(i)}}_{\text{per-patient normalized activity}} + \underbrace{\text{Linear}(\gamma(\tilde{\mathbf{p}}_j))}_{\text{position (constant)}} + \underbrace{\text{sinPE}(t')}_{\text{temporal index}} \quad (3)$$

Temporal binning (shared). $\text{Conv1d}(1 \rightarrow d, k=10, s=10)$: maps each electrode’s scalar HGA to d -dim tokens at non-overlapping 50 ms windows ($200 \text{ Hz} \rightarrow T'$ time bins). Default: $t_{\min}=-0.2$, $t_{\max}=1.0$ ($T'=24$), pending Phase 0 sweep. Shared across all electrodes and patients.

Per-patient diagonal normalization. $s^{(i)} \odot x + b^{(i)}$, $s^{(i)}, b^{(i)} \in \mathbb{R}^d$ (128 params/patient).

Initialized from channel statistics [19]: $s_k^{(i)} = 1/\sigma_k^{(i)}$, $b_k^{(i)} = -\mu_k^{(i)}/\sigma_k^{(i)}$. This starts as z-scoring; the model learns only the residual correction.

Coordinate pipeline (ACPC $\rightarrow$ MNI). Electrode coordinates are stored in ACPC space (per-patient, AC-PC aligned). Before use, they are transformed to MNI-152 via the patient’s `talairach.xfm` affine, then corrected by the learned per-patient rigid calibration:

$$\tilde{\mathbf{p}}_j = R(\omega^{(i)})(T_{\text{xfm}}^{(i)} \mathbf{p}_j^{\text{ACPC}}) + \Delta^{(i)} \quad (4)$$

where $T_{\text{xfm}}^{(i)}$ is the fixed talairach transform (12-param affine, not learned) and $\Delta^{(i)}, \omega^{(i)}$ absorb residual registration error (6 learnable params/patient, hard-clamped $\|\Delta\| \leq 15 \text{ mm}$, $\|\omega\| \leq 0.15 \text{ rad}$, L2-regularized, init at $\mathbf{0}$).

Electrode localization. CT-MRI coregistration $\rightarrow$ grid interpolation $\rightarrow$ cortical surface projection. Error is primarily **systematic** (rigid-body $\sim 4\text{--}8 \text{ mm}$), not random per-electrode ($\sim 0.1 \text{ mm}$) — hence Δ/ω rigid correction. Residual *anatomical* variation ($\sim 15\text{--}25 \text{ mm}$ [4]) is nonlinear and handled by VE cross-attention receptive fields (Section 3.3).

Fourier PE ($F=3$). MNI coordinates $\tilde{\mathbf{p}}_j$ (after talairach transform + learned calibration), standardized so speech motor region spans $[-1, 1]$:

$$\gamma(\bar{\mathbf{p}}) = [\sin(2^0 \pi \bar{\mathbf{p}}), \cos(2^0 \pi \bar{\mathbf{p}}), \dots, \sin(2^{F-1} \pi \bar{\mathbf{p}}), \cos(2^{F-1} \pi \bar{\mathbf{p}})] \in \mathbb{R}^{6F} \quad (5)$$

At $F=3$: wavelengths 30 mm (full-array), 15 mm (articulatory region), 7.5 mm (sub-region). Projected via $\text{Linear}(18 \rightarrow d)$. Fourier PE provides **directional** identity (anisotropic), complementing the distance bias (**proximity**, isotropic). Two equidistant electrodes in different directions are indistinguishable by distance bias alone. **A3** removes PE; **A11** sweeps F .

3.2 Electrode self-attention (ablation **A_elec_attn**)

Not in the default pipeline. The default sends electrode tokens directly to VE cross-attention. Electrode self-attention is an ablation testing whether local spatial preprocessing helps.

Why not default. Electrode self-attention learns spatial relationships *within* each patient’s array. The shared Q/K/V weights operate on tokens containing Fourier PE of MNI coordinates. But the functional meaning of any given MNI location varies substantially across patients (spatial gradient

$R^2 = 0.05\text{--}0.13$; Duraivel 2025). Content-based attention patterns tied to absolute coordinates are patient-specific and unlikely to transfer. The only transferable component is distance-based smoothing (nearby electrodes correlate — physics, not neuroscience), which does not require learned Q/K/V.

A_elec_attn inserts one layer of distance-biased self-attention before VE cross-attention:

$$A_{jk} = \frac{Q_j \cdot K_k}{\sqrt{d_h}} - \alpha_h^{\text{elec}} \|\tilde{\mathbf{p}}_j - \tilde{\mathbf{p}}_k\|^2 \quad (6)$$

$\sim 16\text{K}$ params (4 heads, $d=64$), $O(N_i^2)$ per patient per time bin. If **A_elec_attn** > default, local spatial gradients carry information lost by VE point-sampling; it becomes the new default. **A_elec_attn_no_pe** tests the same layer but with distance bias only (no Fourier PE on electrode tokens), isolating generic spatial smoothing from position-specific patterns.

3.3 VE cross-attention (spatial aggregation)

$L=16$ virtual electrode queries cross-attend to the patient’s N_i electrode tokens, producing a fixed-size $L \times d$ representation per time bin. VE positions are **per-patient**: atlas centroid + learned offset, leashed to prevent drift:

$$\mathbf{p}_l^{(i)} = \mathbf{p}_l^{\text{atlas}} + \delta_l^{(i)}, \quad \|\delta_l^{(i)}\| \leq 15 \text{ mm}, \quad \delta_l^{(i)} \text{ init } \mathbf{0} \quad (7)$$

48 params/patient ($L=16 \times 3$ coordinates), L2-regularized. The atlas provides a strong spatial prior ($N=40$ HCP subjects); the offset adapts to each patient’s individual somatotopy (varies by 5–15 mm). Cross-attention:

$$A_{lj} = \frac{Q_l \cdot K_j}{\sqrt{d_h}} - \alpha_h \|\mathbf{p}_l^{(i)} - \tilde{\mathbf{p}}_j\|^2 \quad (8)$$

The distance bias determines *where* to look (per-patient, via leashed positions). The Q/K/V determines *what* to extract (shared — VE query learns “I am tongue M1, extract tongue-related temporal features” regardless of which patient). This separation is key: spatial selection is per-patient, functional selectivity is shared. At $\alpha_h \approx 0.003$: ~ 25 mm soft receptive fields.

A_no_deform: Fix VE positions at atlas centroids (no per-patient offsets). Tests whether individual somatotopic adaptation matters.

A_dist_only: Remove Q/K/V from cross-attention entirely (distance-only pooling: $w_{lj} = \text{softmax}(-\alpha \|\mathbf{p}_l^{(i)} - \tilde{\mathbf{p}}_j\|^2)$, $z_l = \sum_j w_{lj} x_j$). Tests whether functional selectivity matters beyond spatial proximity.

Per-electrode uncertainty weighting [11]. Each electrode predicts $\sigma_j > 0$ (Linear($d \rightarrow 1$) + softplus, ~ 65 params), modulating cross-attention:

$$A_{lj} = \frac{Q_l \cdot K_j}{\sqrt{d_h}} - \alpha_h \|\mathbf{p}_l^{(i)} - \tilde{\mathbf{p}}_j\|^2 - \beta_h \log \sigma_j^2 \quad (9)$$

VE content vectors. Learned $c_l \in \mathbb{R}^d$ per VE (shared). Queries: $Q_l = W_Q(c_l + \text{FourierPE}(\mathbf{p}_l^{(i)}))$. Fourier PE uses the per-patient VE position, so query embeddings adapt to each patient’s atlas deformation. Initialized from functional profiles: mean HGA response per ROI across reachable patients, projected to d -dim via PCA.

3.4 Spatiotemporal blocks ($\times B$)

After VE cross-attention, the representation is $L \times T'$ tokens ($L=16$ VEs $\times$ 24 time bins). Each block alternates:

(a) **VE spatial self-attention** (per time bin). The 16 VE tokens attend to each other:

$$A_{lm} = \frac{Q_l \cdot K_m}{\sqrt{d_h}} \quad (10)$$

Standard scaled dot-product, 4 heads. Captures inter-regional communication. At $L=16$: 256 entries per time bin.

(b) **Temporal self-attention** (per VE). Each VE attends across 24 time bins. 4 heads, sinusoidal temporal PE.

Factored vs. combined attention. Default factors spatial and temporal. seegificant [8]: factored $>$ joint at electrode-level ($+0.06 R^2$). BarISTA [12]: combined $>$ factored by 1–2 pp at parcel-level, closer to our $L=16$ VEs. At $16 \times 24 = 384$ tokens, cost is trivial either way. **A_combined** tests joint spatiotemporal self-attention.

Block internals. Pre-LN residual connections, ReLU FFN ($d_{ff} = 4d$), dropout $p=0.2$, DropPath $p=0.1$. Channel dropout ($p=0.2$) applied to electrode tokens *before* VE cross-attention as input regularization.

3.5 Virtual electrode atlas positions ($L=16$)

16 left-hemisphere Brainnetome ROI centroids [6], selected as the top 16 by patient reachability (<25 mm) from all 123 LH ROIs. All have ≥ 3 patients within 25 mm. Brainnetome over HCP-MMP1.0 [7] because Brainnetome provides somatotopic motor/sensory subdivisions (tongue M1, face M1, tongue S1) — HCP-MMP’s area 4 centroid is at the dorsal hand region, ~ 30 mm from the ventral face/tongue cortex our arrays cover.

- **Motor** (3): A6cvl (ventral PMC, 9 pts), A4tl (tongue M1, 7 pts), A4hf (face M1, 6 pts)
- **Sensory** (3): A1/2/3tonIa (tongue S1, 8 pts), A1/2/3ulhf (face S1, 5 pts), A2 (proprioceptive S1, 3 pts)
- **Broca’s** (6): A44d (8 pts), A45c (8 pts), A44v (7 pts), A45i (7 pts), A45r (5 pts), A44op (opercular, 4 pts)
- **Auditory** (2): STGpp (planum polare, 4 pts), STGa (anterior STG, 4 pts)
- **Insula** (1): INSa (articulatory planning, 4 pts)
- **Executive** (1): MFG (dorsolateral PFC, 6 pts)

Broca’s-heavy (6/16) because the anterior-ventral electrode cluster (S26, S32, S33, S57) covers IFG well. SMA/pre-SMA unreachable (0 patients, too medial).

Patient-adaptive masking. `select_active_virtual_electrodes()` masks VEs with no electrode within 25 mm. Each patient reaches ~ 6 –12 of 16 VEs. Masked VEs receive zero attention weight, not learned content.

3.6 Learned spatial alternative (ablation)

A_self_attn: Electrode self-attention (no atlas). Replace VE cross-attention with self-attention among N_i electrodes per time bin + Fourier PE. Decoder cross-attends to $N_i \times T'$ tokens (variable size). Follows seegnificant [8].

E2.1: Learned latents (no atlas). Replace atlas positions with $L=16$ learned position vectors. Same Perceiver cross-attention, same bottleneck, everything else identical. Isolates atlas from bottleneck. If learned latents match atlas at $N=10$, the atlas is a convenience. If atlas wins at small N , that’s the core finding.

3.7 Phoneme decoder

Default: lightweight autoregressive heads. Three position-specific heads with AR conditioning — each step sees the previous phoneme:

1. **Temporal pooling per position.** Learned position embeddings $q_k \in \mathbb{R}^d$ ($k=1, 2, 3$) attend to the $T'=24$ time bins via cross-attention (single head): $\text{attn}_k = \text{softmax}(q_k \cdot K/\sqrt{d})$, producing $\bar{\mathbf{z}}_k \in \mathbb{R}^{L \times d}$. Position 1 attends to premotor/onset, position 3 to offset.
2. **Spatial pooling.** Mean-pool over $L=16$ VEs: $\bar{\mathbf{z}}_k \in \mathbb{R}^d$.
3. **AR conditioning.** Concatenate with previous phoneme embedding: $[\bar{\mathbf{z}}_k; \text{emb}(y_{k-1})] \in \mathbb{R}^{2d}$. Step 1 receives a learned BOS token. During training: teacher forcing (ground-truth y_{k-1}). During inference: predicted $\hat{y}_{k-1}$.
4. **Classification.** $\text{Linear}(2d \rightarrow 9) \rightarrow 9\text{-way softmax per position}$.

$\sim 4.1\text{K}$ params total ($16\times$ lighter than a full transformer decoder).

Training:

$$\mathcal{L}_{\text{class}} = \frac{1}{3} \sum_{k=1}^3 \text{FocalCE}(\hat{y}_k, y_k; \gamma=2, \text{LS}=0.1) \quad \text{with mixup } (\alpha=0.2) \quad (11)$$

Inference: beam search with 52-token constraint. Beam search ($B=8$) over the 3-step AR sequence, constrained to 52 valid CVC/VCV tokens ($52/729 = 7.1\%$ of output space).

A_no_ar: remove AR conditioning entirely (independent position heads, no previous phoneme embedding). Tests whether inter-phoneme modeling is worth $\sim 2\text{K}$ params.

3.8 Parameter budget

Capacity: $\sim 175\text{K}$ total ($B=2$), 182/pt (including 48 VE offsets). Fine-tuning on $\sim 1,300$ trials safe via gradual unfreezing (steps 1–2 train $\sim 5\text{K}$ params). **A10** tests $B \in \{1, 2, 3\}$.

4 Training and Evaluation

4.1 Phase 1: Source training

$N-1$ source patients. AdamW (10^{-3}), geometric params (Δ, ω) at 3×10^{-4} . Cosine decay with 20-epoch warmup. Weight decay 10^{-4} , grad clip 5.0, batch 16. Early stopping on 20% held-out source validation, patience 7.

Module	Params	Shared?	Note
Conv1d($1 \rightarrow d, k=10$)	$\sim 0.7K$	Shared	Temporal binning
Fourier PE + Linear (electrodes)	$\sim 1.2K$	Shared	Directional spatial identity
VE content + Fourier PE	$\sim 2.2K$	Shared	$L=16$ learned content + position
VE cross-attention	$\sim 33K$	Shared	Q, K, V, O projections + dist. bias
Blocks ($\times B$)	$\sim 66BK$	Shared	VE self-attn (16×16) + temporal self-attn +
Lightweight AR decoder	$\sim 4.1K$	Shared	Phoneme emb + $3 \times$ (pos attn + Linear($2d \rightarrow$
$s^{(i)}, b^{(i)}$ (diagonal)	128/pt	Per-patient	Signal normalization
$\Delta^{(i)} + \omega^{(i)}$	6/pt	Per-patient	Registration correction
$\delta_l^{(i)}$ (VE offsets)	48/pt	Per-patient	Per-patient VE positions (16×3)
JEPA target enc. (EMA) + predictor (SSL only)	$\sim 16K$	Shared	Predictor: Linear($d \rightarrow 2d$) + GELU + Linear($2d \rightarrow$
AR decoder (A_ar only)	$\sim 67K$	Shared	1-layer transformer + embeddings (ablation)
Shared ($B=2$, supervised)	$\sim 173K$		VE cross-attn 33K + 2 blocks 132K + decoder
Shared ($B=2$, SSL)	$\sim 189K$		+ JEPA predictor 16K (discarded after SSL)
Shared ($B=1$, supervised)	$\sim 107K$		Ablation A10
Per-patient ($\times 10$)	$\sim 1.8K$		182/pt — deliberately lean
Total ($B=2$)	$\sim 175K$		Encoder-heavy: 96% in spatial/temporal proce

4.2 Phase 2: Target adaptation

Per fold of 5-fold grouped-by-token CV on target patient:

Option A: Gradual unfreezing (default). Following the POYO/POSSM/FALCON consensus [17, 18], progressively unfreeze layers instead of binary LP $\rightarrow$ FT. Each step trains until validation loss plateaus (patience 7), then unfreezes the next layer group:

1. **Per-patient calibration.** Freeze all shared weights. Train per-patient params ($s^{(\text{tgt})}, b^{(\text{tgt})}, \Delta^{(\text{tgt})}, \omega^{(\text{tgt})}, \delta_l^{(\text{tgt})}$) only. Target data only. Init $s^{(\text{tgt})}$ from $1/\sigma(\text{channels})$, $b^{(\text{tgt})}$ from $-\mu/\sigma$ (channel statistics; see Section 3.1); δ_l init at **0** (atlas centroid).
2. + **AR decoder.** Unfreeze decoder (phoneme embeddings, position embeddings, cross-attention, output projection). Target data only.
3. + **Last spatiotemporal block.** Unfreeze the final block at $0.1 \times$ LR. **Source replay:** 70% target + 30% source data (uniform across source patients) [20]. Prevents catastrophic forgetting of shared representations.
4. + **All shared weights.** Unfreeze remaining shared weights at $0.1 \times$ LR. Same 70/30 replay. Patience 5 (shorter — diminishing returns from full fine-tuning with limited data).

Option B: Test-time optimization (ablation A_tto). Freeze encoder+decoder. Optimize only per-patient params (182 variables: 128 diagonal + 6 geometric + 48 VE offsets) via L-BFGS on ~ 20 – 50 calibration trials [11]. If the backbone is sufficiently good, TTO converges in seconds and approaches LP-FT — the clinical deployment story.

4.3 Regularization

Default loss:

$$\mathcal{L} = \sum_i w^{(i)} \mathcal{L}_{\text{class}}^{(i)} + \lambda_{\Delta\omega} E_{\Delta\omega} + \lambda_{\text{sm}} E_{\text{smooth}} \quad (12)$$

where $w^{(i)}$ weights each patient by signal quality (sig channel ratio, artifact count; fixed, not learned). Two regularizers active by default:

- $E_{\Delta\omega}$: L2 on geometric params $\Delta^{(i)}, \omega^{(i)}$ ($\lambda=10^{-3}$). Prevents drift to the 15 mm clamp.
- E_{smooth} : Temporal smoothness $\sum_{l,t'} \|\mathbf{z}_{l,t'+1} - \mathbf{z}_{l,t'}\|^2$ (small λ). Neural activity is smooth; prevents learning high-frequency noise.

Plus standard implicit regularization: AdamW weight decay (10^{-4}), dropout ($p=0.2$), DropPath ($p=0.1$), channel dropout ($p=0.2$).

Optional alignment losses (Phase 7). Cross-patient contrastive (same phoneme $\rightarrow$ similar VE reps across patients) and KL distribution matching on VEs [19]. Not default — the architecture should produce aligned VE representations without explicit alignment pressure. Test only if Phase 4 residual analysis shows patient identity dominating the shared space.

4.4 Evaluation

Zero-shot: Phase 1 only, $\Delta=\omega=0$. Pure coordinate generalization.

LP-FT (primary): Phase 1 $\rightarrow$ Phase 2. Matches clinical calibration.

Metrics: PER (edit distance on 3-phoneme sequences), grouped-by-token 5-fold CV, 3 seeds. Full recipe: weighted k -NN ($k=10$), TTA 16, beam search ($B=8$) with 52-token constraint.

4.5 Diagnostics

- **Forward prediction R^2 :** If it stalls as patients are added, heterogeneity is the bottleneck — switch to patient ranking [22].
- **Patient curriculum:** (1) Core 4, (2) +S39, S58, (3) +S23, S16, S22. Stop if validation degrades [15, 22].
- **VE reachability compatibility:** $\text{compat}(i, j) = |\text{activeVEs}(i) \cap \text{activeVEs}(j)| / |\text{activeVEs}(i) \cup \text{activeVEs}(j)|$. Rank source patients for N-curve ordering and SSL selection.

Residual analysis, VE interpretability, variance decomposition: Section 6.5.

5 Self-Supervised Pretraining

Addresses the data bottleneck: $\sim 1,300$ labeled trials vs. 456 min raw continuous EDF (29 patients). Design informed by V-JEPA 2/2.1 [28, 29], BIT [9], BarISTA [12], BrainBERT [13], and RPNT [14].

5.1 SSL objective: JEPA with temporal masking

Design rationale. One objective, not three. V-JEPA 2/2.1 [28, 29] — Meta’s SOTA video SSL — uses a single JEPA loss and achieves state-of-the-art on both understanding and prediction tasks. The video SSL field’s trajectory (pretext tasks $\rightarrow$ contrastive $\rightarrow$ MAE $\rightarrow$ JEPA) shows that intensifying one objective beats diversifying across many. Multi-headed losses (temporal discrimination + swap detection + JEPA) create hyperparameter coupling (λ_{recon} , swap rate, τ , α) that cannot be reliably tuned at our data scale (~ 24 – 100 h). Prior reasoning was: PopT [24] showed discriminative $>$ reconstructive for cross-patient sEEG. But PopT’s task is binary detection — much easier than phoneme sequence decoding, where richer temporal representations are needed. JEPA

learns these by construction: predicting masked temporal content forces the backbone to model articulatory dynamics. L1 loss (V-JEPA 2 recipe) over L2 for robustness to outliers. Content-aware weighting [13] addresses the 68% near-zero problem in z-scored HGA.

5.1.1 JEPA objective

Predict latent representations of masked VE-time positions from visible context via an EMA target encoder ($\tau=0.996 \rightarrow 1.0$). Content-aware weighting ($w_{t'} = 1 + \alpha \|\bar{x}_{t'}\|$, $\alpha \approx 2$ [13]) upweights high-activation bins.

$$\mathcal{L}_{\text{SSL}} = \frac{1}{|\mathcal{M}|} \sum_{t' \in \mathcal{M}} w_{t'} \cdot \|\mathbf{z}_{t'}^{\text{tgt}} - f_{\text{pred}}(\mathbf{z}_{t'}^{\text{online}})\|_1 \quad (13)$$

Masking. Temporal span masking (max 6 patches / 300 ms). Mask ratio $r \sim \text{U}(0, 1)$ per sample [14] — strictly better than any fixed ratio. Optional VE-level spatial masking (mask entire VE positions for some timepoints, forcing cross-VE prediction).

Why JEPA over raw MSE: Latent targets (EMA encoder) provide smoother learning signal than pixel-space reconstruction. BarISTA [12] confirmed for iEEG. BrainBERT [13] showed content-aware weighting is critical for sparse neural signals.

Why not multi-headed SSL: Electrode-level swap detection (prior design) is architecturally misaligned — it trains representations at electrode level before VE cross-attention, but the default architecture has no electrode self-attention. The gradient doesn’t reach VE or temporal layers. Temporal discrimination is a pretext task analogous to “predict frame order” in video — V-JEPA showed these are superseded by masked prediction. Both are tested as ablations (A_disc, A_swap).

5.1.2 Dense prediction loss (ablation A_dense)

Inspired by V-JEPA 2.1 [29]: supervise *visible* (unmasked) tokens in addition to masked tokens, with distance-weighted supervision near mask boundaries:

$$\mathcal{L}_{\text{dense}} = \mathcal{L}_{\text{SSL}} + \frac{1}{|C|} \sum_{i \in C} \frac{\lambda}{\sqrt{d_{\min}(i, \mathcal{M})}} \cdot \|\mathbf{z}_i^{\text{tgt}} - f_{\text{pred}}(\mathbf{z}_i^{\text{online}})\|_1 \quad (14)$$

This prevents context tokens from degenerating into global aggregators. **Caution:** V-JEPA 2.1 found context loss harms classification (−10 pp) unless paired with deep supervision across intermediate encoder layers. With only $B=2$ blocks, recovery may be limited. Not default — test as ablation.

If audio annotations are available from external data: add neural-audio contrastive (CLIP-style InfoNCE [21]) as auxiliary. One-sentence future direction, not default.

Validation. Downstream linear probe accuracy on a small held-out labeled set for checkpoint selection, not reconstruction R^2 .

5.2 Per-patient layers during SSL

Following BIT [9], per-patient diagonal normalization is learned during SSL; the backbone, target encoder, and predictor are shared.

Component	During SSL	During supervised
Shared Conv1d + per-patient $s^{(i)}, b^{(i)}$	Same as supervised	Same
Fourier PE + Linear	Shared	Same
VE cross-attn + spatiotemporal blocks	Shared (encoder)	Same
JEPA target encoder (EMA) + predictor	Shared ($\sim 16K$)	Removed
Autoregressive decoder	Not used	Shared (init from scratch)
$\Delta^{(i)}, \omega^{(i)}$	Learned during SSL	Transferred or reset

5.3 Data augmentation

Signal-domain (BIT recipe [9] for SSL; lighter for supervised): white noise SD = 0.10–0.20, constant offset SD = 0.05–0.10, amplitude scaling SD = 0.15–0.25, Gaussian smoothing ($w=2.0$), channel dropout $p=0.1$. No time shift for continuous SSL data (random windowing is implicit).

Coordinate-domain [11]: coordinate jitter ($\sigma=2\text{--}5$ mm), global rigid perturbation ($\Delta \leq 5$ mm), electrode subset sampling (70–100%), VE atlas position jitter ($\sigma=3\text{--}8$ mm). Teaches robust receptive fields against registration error and anatomical variation.

5.4 SSL training recipe

Primary pipeline (2-stage, contingent on external data access).

1. **Stage 1: External chronic ECoG SSL (JEPA).** Large-scale pretraining on chronic epilepsy-monitoring ECoG during speech production. Target sources: Flinker/Chen lab, NYU (48 patients, upon request [25]) and Chang lab, UCSF ($\sim 15\text{--}25$ patients, upon request [26]). Estimated $\sim 50\text{--}100$ h of diverse speech (sentence reading, word naming, CV syllables). SSL is task-agnostic — JEPA uses temporal masking, not labels. Per-patient diagonal normalization learned during SSL; backbone shared. Chronic recordings provide rich, varied temporal dynamics (days of naturalistic monitoring), unlike intra-operative data (~ 30 min, repetitive task).
2. **Stage 2: Per-patient supervised fine-tune on CoganLab.** Encoder initialized from Stage 1 (discard JEPA target encoder + predictor). AR decoder trained from scratch. Gradual unfreezing (default) vs. LP-only vs. full FT (Phase 3 comparison). 46–178 trials per patient. This IS the clinical calibration step: pretrained model $\rightarrow$ 20–50 trials $\rightarrow$ decoding.

Why SSL over supervised pretraining: External data has different speech tasks (sentence reading, CV syllables) with different phoneme inventories. Supervised pretraining would require label alignment across datasets. SSL sidesteps this — it learns general spatiotemporal representations from any speech-related neural activity. The 9-phoneme CVC/VCV head trains only on CoganLab data.

Why not sEEG SSL: CoganLab sEEG (~ 1000 min, 25 patients) is same-modality only in the sense that both measure field potentials. But sEEG motor cortex coverage is $\sim 15\text{--}30\%$ of patients (clinical trajectories target seizure zones, not speech areas), electrode density is sparse (~ 10 contacts per region), and transfer from depth to surface recordings is unproven. If external chronic ECoG is available, sEEG pretraining adds marginal value at best.

Fallback pipeline (3-stage, CoganLab-only). If external data is not obtainable:

1. sEEG SSL (~ 1000 min, 25 patients, *if motor VE coverage ≥ 3 patients* — Phase 0 check).

2. μ ECoG SSL (456 min, 29 patients, $\sim 52\%$ task cycles; content-aware weighting + segment sampling).
3. Supervised fine-tune.

Total ~ 24 h. Adequate but limited by intra-operative data quality and sEEG coverage gaps.

Patient ranking. Do not blindly pool [22, 15]. Rank by VE coverage similarity and HGA quality. **A_rank** ablates.

Training. AdamW, $\text{lr}=5 \times 10^{-4}$, weight decay 10^{-5} , batch 64, cosine decay, bfloat16. Checkpoint: best downstream linear probe accuracy.

Transfer. Stage 1 $\rightarrow$ 2: discard JEPA target encoder + predictor; transfer encoder (VE cross-attn, spatiotemporal blocks) + shared layers; per-patient params init from channel statistics; AR decoder from scratch.

6 Experimental Design

Phased design: each phase answers one question and gates the next. ~ 44 experiments, ~ 110 training runs at 3 seeds on key comparisons. All use grouped-by-token 5-fold CV, 3 seeds, core 4 patients unless noted. Every ablation changes **exactly one thing** from the default.

6.1 Phase 0: Sanity + temporal window

6.1.1 Prerequisite: nonlinear MNI normalization

Hard blocker. Current coordinates use `talairach.xfm` (12-param linear affine), leaving $\sim 5\text{--}15$ mm of nonlinear residual that Δ/ω (rigid, 6 params) cannot correct. Every spatial mechanism — Fourier PE, distance bias, `A_deform`, `A_spatial_flow` — operates on these coordinates. If T1 structural MRIs exist for each patient, ANTs SyN nonlinear normalization to MNI-152 could halve coordinate error from 15–25 mm to 5–10 mm. **Action:** confirm T1 availability with Zac; run ANTs if available; re-evaluate coordinate quality before committing to Phase 0 experiments.

6.1.2 Temporal window sweep (per-patient v12, S14)

The full-trial AR decoder operates over the entire input window — the wrong window could tank Phase 0 before the architecture is tested. Per-phoneme sweeps found $t_{\min} = -0.15$ optimal for single-phoneme epochs, but full-trial dynamics differ (3 phonemes across ~ 450 ms, position-specific attention, premotor planning in Broca’s). Sweep $t_{\min}$ first at fixed $t_{\max} = 1.0$, then $t_{\max}$ at best $t_{\min}$:

Sweep	Values
$t_{\min}$ (fixed $t_{\max} = 1.0$)	$\{-0.5, -0.3, -0.2, -0.1, 0.0\}$ s
$t_{\max}$ (fixed best $t_{\min}$)	$\{0.6, 0.8, 1.0\}$ s

~ 8 runs on per-patient v12, S14 only (fast). Timing context: stimulus-to-response delay 1.1 ± 0.3 s, stimulus duration ~ 500 ms [1], utterance ~ 450 ms. $t_{\min} = -0.5$ reaches into the post-stimulus gap; $t_{\min} = -0.2$ captures premotor planning; $t_{\min} = 0.0$ is production-only. $t_{\max} = 0.6$ covers the full utterance; $t_{\max} = 1.0$ adds ~ 550 ms post-utterance. Winner becomes the default for all subsequent experiments.

6.1.3 Cross-patient sanity check

Experiment	What it answers
Per-patient Conv2d+BiGRU baseline	Anchor (done: PER ~ 0.734 S14, ~ 0.825 pop mean)
Cross-patient v12 default (LOPO, core 4)	Does v12 cross-patient beat per-patient baselines?

Gate: v12 $\geq$ baseline on $\geq 2/4$ patients $\rightarrow$ proceed. Else debug spatial mechanism.

6.1.4 sEEG coverage inventory (gates Phase 6)

Before building the sEEG SSL pipeline, compute VE reachability for CoganLab sEEG electrode positions using existing `atlas.py` tools. Clinical sEEG is implanted for seizure localization, not speech research. Coverage estimates: STG $\sim 50\text{--}70\%$ of patients, IFG $\sim 30\text{--}50\%$, **ventral precentral gyrus** $\sim 15\text{--}30\%$. Motor cortex is the most speech-relevant and least likely to be covered.

Requirement: ≥ 3 sEEG patients must reach each motor VE (A6cv1, A4tl, A4hf) at 25 mm threshold. If not, motor VEs will receive no useful gradient during sEEG SSL, and the Stage 1 value proposition weakens to auditory/IFG pretraining only. **Fallback:** skip Stage 1, go directly to μ ECoG SSL $\rightarrow$ supervised.

6.2 Phase 1: Scaling

Question: Does performance improve as patients and data are added?

6.2.1 Patient N -scaling curve

Train on $N \in \{1, 2, 4, 7, 10\}$ source patients, evaluate on target via LOPO. Source ordering by VE reachability compatibility. $N=1$ is per-patient v12 — the bottom of the curve, not a pass/fail gate.

Experiment	What it answers
N-curve, target S14	Scaling on best patient (posterior-dorsal cluster)
N-curve, target S26	Scaling on anterior-ventral cluster
N-curve, target S33	Stress test (46 trials, smallest patient)
N-curve, random ordering	Does patient ranking matter? (Jiang: $8\times$ efficiency [22])

~ 15 runs ($5 N$ values $\times 3$ targets). 3 seeds on $N \in \{1, 4, 10\}$. **Expected:** log-linear improvement, plateau $N=7\text{--}10$. Flat/decreasing $\rightarrow$ foundation model story is dead.

Key secondary: Run E2.1 (learned latents) across the N-curve. If atlas $>$ learned at small N but $\approx$ at large N : “atlas prior most valuable in small- N regime.”

6.2.2 sEEG cross-modality transfer

Tests the modality-agnostic claim. **Gated on sEEG data inventory + HGA extraction.**

Paper impact: If sEEG helps, data ceiling goes from ~ 8 h to 1000s h. No prior sEEG $\rightarrow \mu$ ECoG speech transfer.

Experiment	What it answers
sEEG+ μ ECoG SSL $\rightarrow$ supervised	Does sEEG improve μ ECoG decoding?
sEEG-only SSL $\rightarrow$ μ ECoG supervised	Can sEEG alone pretrain for μ ECoG?
sEEG supervised transfer (if labels exist)	Direct cross-modality without SSL

6.3 Phase 2: Mechanism — controlled ablations

Each changes exactly one component from the default.

6.3.1 Spatial mechanism (paper’s core claim)

ID	Change	What it isolates
E2.1	Learned latents ($L=16$, no atlas)	Atlas prior vs. any Perceiver bottleneck. Key ablation.
A_self_attn	Electrode self-attn (no VEs)	Bottleneck itself: is $N_i \rightarrow L$ compression necessary?
A_no_dist	Remove distance bias	Content-only cross-attn. Is proximity weighting necessary?
A3	Remove Fourier PE	Distance bias only. Is directional encoding needed?
A_no_deform	Fix VE positions at atlas centroids (no per-patient offsets)	Is per-patient somatotopic adaptation necessary?
A_dist_only	Remove Q/K/V from cross-attn (distance-only pooling)	Is functional selectivity needed beyond spatial proximity?
A_elec_attn	Add electrode self-attn before VE cross-attn	Does local spatial preprocessing help?
A_elec_attn_no_pe	Electrode self-attn, distance bias only (no PE)	Generic spatial smoothing vs. position-specific patterns

Core comparison: E2.1 vs. default disentangles atlas from bottleneck (see Section 3.6).

Per-patient VE positions (default). Brainnetome centroids are population averages ($N=40$); individual somatotopy varies by 5–15 mm [3, 4]. The motor VE gap (A4t1 $\leftrightarrow$ A4hf: 32 mm) exceeds the full dorsal-ventral articulatory gradient (~ 15 –30 mm). Per-patient learnable 3D offsets (+48 params/pt) let each VE shift toward the patient’s true functional anatomy, leashed at 15 mm. The atlas initialization provides a strong prior; the offsets handle individual variation. **A_no_deform** fixes VE positions at atlas centroids, testing whether individual adaptation matters.

Distance-only pooling (A_dist_only). The default VE cross-attention uses Q/K/V to learn *what* to extract from nearby electrodes (functional selectivity) plus distance bias for *where* to look. A_dist_only removes Q/K/V entirely: $w_{lj} = \text{softmax}(-\alpha \|\mathbf{p}_l^{(i)} - \tilde{\mathbf{p}}_j\|^2)$, $z_l = \sum_j w_{lj} x_j$. Tests whether functional selectivity matters — 61% of responsive electrodes encode multiple articulators [27], so selective attention may help the VE extract its target articulator from mixed signals.

Electrode self-attention (A_elec_attn). Inserts self-attention before VE cross-attention. Shared Q/K/V operate on Fourier PE of MNI coordinates, whose functional meaning varies across patients ($R^2 = 0.05$ –0.13). Content-based spatial patterns are likely patient-specific. **A_elec_attn_no_pe** tests distance-bias-only electrode self-attention (no PE), isolating generic spatial smoothing.

6.3.2 Negative controls (mandatory sanity checks)

If any of these pass (match default), the corresponding claim is dead. Run early — before investing in scaling or SSL.

ID	Change	What it kills
A_shuffle_coord	Permute electrode MNI coords within patient	Coordinates carry no spatial information
A_random_atlas	Random VE positions (coverage-matched)	Atlas specificity is irrelevant
A_no_auditory	Remove auditory VEs (STGpp, STGa)	Model exploits auditory feedback, not motor execution
A_late_window	$t_{\min}=0.2$ s (production only)	Model relies on premotor planning, not execution

Auditory confound. The auditory stimulus ends ~ 600 ms before response onset (1.1 ± 0.3 s stimulus-to-response delay, ~ 500 ms stimulus duration [1]), so the default input window ($t_{\min} = -0.2$ s) does not overlap with the stimulus. The remaining confound is **auditory self-monitoring**: auditory VEs (STGpp, STGa) could decode from the patient hearing their own speech during production, not from motor execution. A_no_auditory removes these VEs; A_late_window starts at $t=0.2$ s (well into production, past any premotor transient). If neither hurts, motor cortex is doing the work.

Coordinate corruption. A_shuffle_coord randomly permutes electrode MNI coordinates across electrodes within each patient — same activity, same electrode count, wrong spatial identity. If performance is unaffected, coordinates provide no information and the entire spatial claim collapses. A_random_atlas replaces the 16 Brainnetome atlas positions with 16 random left-hemisphere cortical positions (matched for mean electrode coverage). If random $\approx$ atlas, the specific positions don't matter — the Perceiver bottleneck alone explains the result.

6.3.3 Per-patient calibration

ID	Change	What it isolates
A15	Remove diagonal (s, b)	Only Δ/ω (6 params/pt). Is impedance calibration needed?
E2.6	Remove ALL per-patient params	Zero-shot (PopT style). Pure coordinate generalization.

Two experiments bracket the range. No intermediate variants needed.

6.3.4 Decoder

ID	Change	What it isolates
A_no_ar	Remove AR conditioning	Independent heads. Is phoneme-to-phoneme modeling worth $\sim 2K$?

6.4 Phase 3: Adaptation protocol

Question: How should we adapt to new patients, and how much data is needed?

Protocol	What it tests
Linear probe only	Freeze backbone. Train per-patient + decoder only. Minimal forgetting risk.
Full fine-tune	Unfreeze all, same LR. Max flexibility, may overfit.
Gradual unfreezing (default)	Progressive unfreeze, $0.1\times$ LR. Is the complexity justified?

6.4.1 Training protocol comparison

These three bracket the adaptation space.

6.4.2 Calibration efficiency curve

Train on $N-1$ source patients, adapt to target with $\{5, 10, 20, 50, 100, \text{all}\}$ trials. Plot PER vs. calibration data for each target patient.

- **Clinical:** If plateau at 20 trials (~ 30 s), a single calibration sentence suffices.
- **SSL interaction:** Re-run after Phase 6. If SSL shifts the curve left (fewer trials needed), that’s the practical SSL benefit.
- **Comparison:** Both v12 and Conv2d+BiGRU baselines. Earlier plateau for v12 = better initialization from shared backbone.

6.5 Phase 4: Diagnostics & interpretability

Analyses on existing runs — no new training. Validates whether the architecture achieves its stated goals.

6.5.1 Per-patient residual analysis

The architecture claims calibration produces a patient-invariant shared space. Measure:

1. **Diagonal magnitude.** Histogram of $|s_k^{(i)} - 1|$ across features and patients. Small (< 0.5) = channel-statistics init works, minimal correction needed. Large = heavy per-patient correction, possible information leakage into shared space.
2. **Δ/ω magnitude.** $\|\Delta^{(i)}\|$ and $\|\omega^{(i)}\|$ per patient. Compare to known registration error ($\sim 4-8$ mm). If Δ hits the 15 mm clamp consistently, clamping is too tight or coordinates are bad.
3. **Variance decomposition.** ANOVA on VE representations: fraction of variance from phoneme identity (signal) vs. patient identity (should be small) vs. sequence position. If patient identity $> 50\%$, calibration is insufficient.
4. **Cross-patient CKA.** For same phoneme, CKA between VE reps across patient pairs. High = patient-invariant space achieved. Low = per-patient effects dominate after calibration.

6.5.2 VE interpretability

1. **VE activation profiles.** Mean activation per VE per phoneme class across time. Do motor VEs (A4tl, A4hf) activate during production? Auditory VEs (STGpp, STGa) during perception?

Broca’s (A44/A45) pre-motor (phonological assembly)?

2. **Region discriminability probe.** Linear probe: VE outputs $\rightarrow$ 16-way VE index. Discriminative = spatial mechanism preserves regional identity. Uniform = VE cross-attention collapsing [16].
3. **Phoneme selectivity.** Which VEs discriminate which phonemes? Does tongue M1 (A4tl) preferentially encode tongue phonemes (/t/, /k/, /g/) over lip phonemes (/b/, /p/, /v/)?
4. **Attention weight maps.** Per patient: which real electrodes contribute to which VEs? Do patterns respect anatomy (ventral $\rightarrow$ motor VEs, superior $\rightarrow$ sensory)?
5. **VE trajectory analysis.** PCA on VE reps across time bins. 3-phoneme tokens trace distinct paths? Similar phonemes cluster at the relevant sequence position?

6.6 Phase 5: Hyperparameter sensitivity

ID	Sweep	What it answers
A_dim	$d \in \{32, 64, 128\}$	Hidden dimension. Novel: no prior iEEG paper ablates d .
A10	$B \in \{1, 2, 3\}$	Block depth (default $B=2$)
A_thresh	VE threshold $\in \{15, 25\}$ mm	Receptive field vs. coverage

~ 7 runs on default config, core 4 patients.

6.7 Phase 6: SSL

Hard dependency: external data access (Flinker/Chang) for primary pipeline; HGA extraction for fallback.

6.7.1 SSL recipe

ID	Experiment	What it answers
A18	SSL $\rightarrow$ FT vs. from scratch	Does SSL help at all?
A_mse	Raw L2 MSE replacing JEPA L1	Are latent targets worth the EMA complexity?
A_dense	Dense prediction loss (V-JEPA 2.1 [29])	Does supervising visible tokens help?
A_deep	+ hierarchical supervision (3 levels)	Does deep supervision recover classification?
A_disc	+ temporal discrimination auxiliary	Does discriminative add value over JEPA?
A_spatial_mask	+ VE-level spatial masking	Does combined masking help?
A_rank	Ranked vs. uniform pooling	Does Jiang-style selection matter?
A_domain	+CoganLab μ ECOG domain adapt	Does intermediate in-domain SSL help?

A_dense adds V-JEPA 2.1’s context loss: supervise unmasked tokens with distance-weighted weighting ($\lambda_i = \lambda/\sqrt{d_{\min}(i, \mathcal{M})}$). Improves dense/spatial representations but harms classification (-10 pp) without deep supervision [29]. A_deep adds hierarchical supervision at 3 levels (post-VE-cross-attn, post-block-1, post-block-2) to recover. A_domain tests whether adding a CoganLab μ ECOG SSL stage between external pretraining and supervised fine-tuning improves over direct fine-tuning.

6.7.2 SSL scaling analysis

SSL on $\{1, 2, 4, 8, 16, 50, 100\}$ h of external data, evaluate downstream supervised PER.

- Is scaling log-linear? Where does it plateau? BIT: 20–30 h productive range.
- Compare external ECoG only vs. external + CoganLab at each scale point. If CoganLab in-domain data shifts the curve up, domain adaptation is justified.
- **Cross-density transfer:** SSL on ~ 10 mm (Flinker) vs. ~ 4 mm (Chang) vs. mixed. Does electrode density matter for SSL, or does the VE bottleneck abstract it away?
- **Interaction with calibration:** Re-run calibration curve (Phase 3) with SSL. If SSL reduces trials needed (e.g., $50 \rightarrow 10$), that’s the practical foundation model benefit.

6.8 Phase 7: Refinements (conditional)

Run only if Phases 1–6 justify:

Experiment	What it tests	Trigger
Full-trial vs. per-phoneme input	Does per-phoneme give same ~ 6 pp boost?	v12 works
Cross-task data (with/without Duraivel)	Validate default inclusion	Before population eval
Joint spatiotemporal attention	Combined $L \times T'$ vs. factored	Factored results strong
KL distribution matching on VEs	NoMAD-style alignment [19]	Residuals large (Phase 4)
Test-time optimization	L-BFGS on per-patient only	Clinical deployment

6.9 Experiment summary

Phase	Expts	Runs	Gate
0: Sanity + window + sEEG inventory	3	~ 14	Window selected; $\geq 2/4$ patients; sEEG coverage verified
1: Scaling	7	~ 20	Performance increases with N
2: Mechanism	16	~ 32	Ablations + negative controls + A_no_deform + A_dist_only + .
3: Adaptation	4	~ 12	Protocol + calibration curve
4: Diagnostics	0	0	Analysis on existing runs
5: Hyperparams	3	~ 7	Sensitivity
6: SSL	8	~ 19	SSL > scratch
7: Refinements	5	~ 10	Conditional
Total	~ 43	~ 108	

6.10 Success criteria

Primary: Cross-patient v12 PER ≥ 3 pp over per-patient Conv2d+BiGRU on $\geq 3/4$ core patients. Paired across patients $\times$ seeds ($n=12$), Wilcoxon $\alpha=0.05$.

Scaling: Positive slope in N-curve. PER improves monotonically $N=1 \rightarrow 10$.

Mechanistic: Atlas > learned latents (E2.1) by ≥ 2 pp at $N=4$. VE variance decomposition: phoneme > patient identity.

Clinical: Calibration curve plateaus within 50 trials (~ 1 min).

6.11 Risks

Risk	Sev.	Mitigation / fallback
$v_{12} \leq$ per-patient baseline	High	Debug spatial; fallback <code>A_self_attn</code>
N-curve flat or decreasing	High	Patient ranking; reduce per-patient capacity; check residuals
Atlas $\approx$ learned latents	Med	Paper becomes “Perceiver bottleneck” (still publishable)
Residuals dominate shared space	Med	Escalate per-patient capacity; nonlinear MNI
Atlas misaligned (talairach.xfm)	High	Phase 0 prerequisite: ANTs SyN if T1s exist; else $\Delta/\omega + A_deform$
175K overfits 1,300 trials	Med	LP-only; dropout; $B=1$; SSL pretraining
sEEG transfer fails	Med	μ ECoG-only SSL (~ 7.6 h); lower data ceiling
μ ECoG SSL low-content ($\sim 48\%$ overhead)	Low	Content-aware weighting + segment sampling; fallback drop Stage 2
HGA extraction blocked	High	Must implement EDF $\rightarrow$ HGA before SSL
JEPA EMA collapse	Low	Monitor target diversity; fallback raw MSE

7 Available Data

Cohort	Patients	Trials	MNI?	Usage
Core set (Zac-recommended)	S14, S26, S33, S62	~ 609	Yes	Source / target
Additional	S23 [†] , S39, S58 [‡]	~ 430	Yes	Source
Cross-task (Duraivel 2025)	3 overlapping [¶]	156–208/pt	Expected	Source (default)
Excluded	S32 [§] , S57	—	—	Excluded
Excluded (poor coverage)	S16 [#] , S22 ^{**}	—	—	Excluded

[†]S23: flagged as noisy by Zac. [‡]S58: right hemisphere, motor & IFG coverage. [§]S32: no HG response (per Zac’s quality spreadsheet) + missing sigChannel.mat. ^{||}S57: only 52/256 significant channels + hybrid strip array. [#]S16: array slid anterior from motor cortex toward IFG — explains poor decoding. ^{**}S22: right hemisphere, poor MFA alignment. [¶]Cross-task: same 9 phonemes, same CVC/VCV task, same IRB/pipeline [2]. 3 patients overlap with our cohort. Nearly doubles training data for those patients.

Patient selection. Core set prioritized by Zac for HGA quality and motor cortex coverage. S23, S39, S58 usable but noisier. S32/S57 excluded for quality; S16 for coverage (IFG, not motor). Cross-task data included by default (see footnote [¶]).

7.1 Pretraining corpus

Any intracranial field potential recording (HGA + MNI coordinates) is valid pretraining data. Utah array spike data is excluded (different biophysics).

Primary plan: external chronic ECoG (Flinker + Chang, est. ~ 50 – 100 h, ~ 60 – 70 patients). **Fallback:** CoganLab-only (~ 24 h, sEEG + μ ECoG). Bouchard/Chang Figshare data (3 patients, public) available immediately for pipeline validation.

Source	Modality	Hours	Patients	Status
Flinker/Chen lab (NYU)	ECoG	est. 50+	48	Request (PI contact)
Chang lab (UCSF)	ECoG	est. 20+	~15–25	Request (PI contact)
Bouchard/Chang (Figshare)	ECoG (256-ch)	small	3	Public (NWB)
CoganLab μ ECoG (continuous)	uECoG	~7.6	29	Have EDF, need HGA extraction
CoganLab sEEG (speech)	sEEG	~16.7	25	On Box (fallback only)

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